

# HEMANT KUMAR JAISWAL

Electrochemical Engineer | Battery R&D & Analytics | Techno-Economics | Product Ownership

New Delhi, India | +91-8756643102 | hkjaiswal7@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

## PROFESSIONAL SUMMARY

Electrochemical engineer with 5+ years of combined industry and research experience across battery systems, fuel cells/electrolysers, functional materials, process scale-up and data analytics. Hands-on experience spans electrode/cathode development, EIS/CV/LSV, cell and stack testing, degradation and failure analysis, materials characterisation, pilot-scale process engineering and 20 kW Metal-Air product development. Built portfolio-grade battery and SOFC analytics workflows using Python, statistical/ML methods and physics-aware interpretation. Experience also includes CAPEX/OPEX modelling, LCOS/LCOH, clean-energy feasibility studies and technology-readiness assessment. Known for taking ownership from problem definition and experimentation through vendor coordination, manufacturing handover, technical reporting and commercial decision support.

## CORE COMPETENCIES

**Electrochemistry & R&D:** EIS | CV | LSV | Chronoamperometry | Polarisation | Electrode/MEA Development | Degradation & Failure Analysis | DOE | 8D | 5-Why

**Battery & Energy Systems:** Metal-Air | Li-ion | Na-ion | Redox Flow | Solid-State Fundamentals | SOFC/PEMFC | AWE/AEM | Cell/Stack Testing | SOC/SOH/RUL | Thermal & Impedance Diagnostics

**Battery Analytics:** Python | Pandas | NumPy | SciPy | Scikit-learn | XGBoost | Time-Series Analysis | Feature Engineering | Regression | Forecasting | Validation | Uncertainty | PyBaMM

**Process & Commercial:** Process Scale-Up | TRL/MRL | Manufacturing Readiness | PFD/P&ID | Aspen Plus | CAPEX/OPEX | LCOS/LCOH | Sensitivity Analysis | TEA | Vendor/BOM Management

**Characterisation & Tools:** SEM-EDS | XPS | XRD | FTIR | BET | TGA/DSC | HRTEM | ICP-OES | MATLAB/Simulink | Git/GitHub | PostgreSQL

## PROFESSIONAL EXPERIENCE

**Chakr Innovation Pvt. Ltd. | Senior Engineer - Electrochemical Systems & Product Development** | Gurugram | Jan 2025 - Apr 2026

- Owned electrochemical development and troubleshooting for a 20 kW Metal-Air battery product, working from cathode preparation and resistance control through cell testing, stack assembly, leak/pressure validation and system performance.
- Implemented in-plane and through-plane cathode resistance measurements, reducing test turnaround from ~2 days to ~2 hours and supporting >50% reduction in cathode resistance through process optimisation.
- Diagnosed recurring performance failures using EIS, cycling trends and post-mortem SEM/XPS/FTIR; translated root causes into material, electrolyte and process corrective actions.
- Delivered ~75% material-cost reduction and ~75% process-yield improvement through first-principles redesign, structured experimentation and production-linked R&D.
- Prepared test plans, SOPs, BOMs and production plans; coordinated suppliers and custom components, supported manufacturing handover, and presented battery-system methodology and results to OEM stakeholders including Maruti Suzuki.
- Contributed to TRL/MRL, manufacturing-readiness and scale-up risk assessments, balancing electrochemical performance, manufacturability, cost and commercial viability.

**Indian Institute of Technology Delhi | Researcher Scholar- Electrochemistry & Energy Systems** | New Delhi | Jul 2023 - Dec 2024

- Developed and tested alkaline/AEM water-electrolysis materials and assemblies, including membrane synthesis/characterisation, electrode preparation, MEA fabrication, stack assembly and electrochemical evaluation.
- Used EIS, polarisation, HER/OER overpotential and Tafel analysis to connect material/interface behaviour with stack performance and degradation.
- Designed an electrolyte-recovery process using RO, lamella clarification and crystallisation; developed PFD/P&ID, material-balance and equipment/RFQ inputs for scale-up.
- Completed advanced energy-storage coursework covering Li-ion, Na-ion, redox-flow, metal-air and solid-state battery concepts, state estimation, degradation, thermal behaviour and reliability.

**CSIR - National Chemical Laboratory | Project Associate I** | Pune | Oct 2022 - Jul 2023

- Synthesised 2-5 nm Pt/C electrocatalysts and characterised structure, surface chemistry and composition using HRTEM, XRD, XPS, BET, FTIR, SEM-EDS and ICP-OES.
- Fabricated PEM MEAs using catalyst-ink preparation, spray/decal coating and hot pressing; evaluated electrochemical performance using CV, LSV, EIS and ECSA.
- Designed and validated continuous-flow/pilot processes, including reactor sizing, control logic, process optimisation and scale-up, with publication in Reaction Chemistry & Engineering (2024).

## SELECTED ANALYTICS & TECHNOLOGY PROJECTS

---

### 1. Battery Fleet Intelligence - Multi-Year Residential Storage Systems

- Developed a reusable Python analytics architecture for multi-year field telemetry covering ingestion/quality, operating regimes, event detection, KPIs, degradation/SOH, anomaly root-cause analysis, forecasting/RUL, ECM/EKF state estimation, dispatch optimisation and uncertainty/explainability.
- Designed validation safeguards for data leakage, pseudo-replication, weak ground truth, distribution shift, sensor drift, battery-to-battery dependence and uncertainty estimation; structured the pipeline for expansion from pilot systems to the full fleet.

### 2. SOFC Health Intelligence - Multimodal Time-Series & EIS Analytics

- Built a physics-aware workflow across EIS, I-V and transient data for multiple SOFC cells, integrating data audit, SOH target design, causal feature engineering, rolling-origin validation and multimodal health forecasting.
- Engineered reusable electrochemical/time-series features and used cross-cell analysis to distinguish performance, impedance and degradation signatures while preserving battery/cell-level independence.

### 3. NASA Battery Aging - SOH & RUL Analytics

- Built a Python pipeline for cycle-level battery degradation analysis; quantified 28.62% capacity loss, achieved  $\sim 0.98 R^2$  for SOH regression and developed a final RUL workflow with 14.79-cycle MAE and 14.85-cycle RMSE.
- Benchmarked regression and tree-based approaches investigated model/split failures and maintained reproducible project structure with Git/GitHub.

## TECHNO-ECONOMIC, FINANCIAL & TECHNOLOGY ASSESSMENT

---

- Built techno-economic and financial models for Metal-Air battery, compressed biogas and bioethanol projects, including CAPEX/OPEX, revenue/cash-flow logic, sensitivity analysis, commercial feasibility and investment-oriented decision support.
- Performed LCOS and LCOH calculations for energy-storage and hydrogen/electrolysis assessments, connecting engineering assumptions with cost and commercial metrics.
- Prepared management-level policy briefs on SATAT, GOBARdhan and PM-JIVAN and developed working knowledge of critical-mineral and clean-energy supply-chain issues.

## EDUCATION, CERTIFICATIONS & PUBLICATION

---

M.Tech, Chemical Engineering - CSIR-NCL Pune

B.Tech, Chemical Engineering - IET Lucknow

**Certifications:** Aspen Plus Advanced (NPTEL) | Machine Learning for Core Engineering Disciplines (NPTEL) | Fundamentals of Electronic Materials & Devices (NPTEL)

**Publication:** Jundale et al. (2024), Reaction Chemistry & Engineering, 9, 1914-1923 - continuous-flow process scale-up research.

**Recognition:** Fast-track promotion to Senior Engineer within 11 months | Value Champion of the Month - Chakr Innovation